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Day 11

CSS

Filters and Blend Modes

Filter

It is a popular CSS tool that is used to:

- to apply graphic effects like blur, brightness, contrast, etc on images, backgrounds, text, and even videos

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Common Filters

Function	Use Case	Example Code
`blur(px)`	Softens the element or background	`filter: blur(5px);`
`brightness(%)`	Lightens or darkens an element	`filter: brightness(70%);`
`contrast(%)`	Increases contrast	`filter: contrast(150%);`
`grayscale(%)`	Converts to black and white	`filter: grayscale(100%);`
`sepia(%)`	Applies a warm tone	`filter: sepia(80%);`
`saturate(%)`	Intensifies colors	`filter: saturate(200%);`
`hue-rotate(deg)`	Shifts color hues	`filter: hue-rotate(90deg);`
`invert(%)`	Inverts colors (good for dark mode)	`filter: invert(100%);`
`drop-shadow(...)`	Adds shadow that follows shape	`filter: drop-shadow(2px 4px 6px black);`

Backdrop Filters

It applies effects to the background behind the element (like frosted glass).

Function	Use Case	Example Code
`blur(px)`	Creates frosted glass look	`backdrop-filter: blur(10px);`
`brightness(%)`	Highlights blurred background	`backdrop-filter: brightness(1.2);`
`contrast(%)`	Enhances contrast behind element	`backdrop-filter: contrast(150%);`
`grayscale(%)`	Mutes the background color	`backdrop-filter: grayscale(1);`

> ✅ Don't forget to use `background-color: rgba()` for semi-transparency when using `backdrop-filter`.

Mix Blend Mode

It defines how an element blends with what's beneath it.

It is similar to Photoshop layer effects and common among graphic designers.

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Blend Modes

Function	Use Case	Example Code
`normal`	Default, no blending	`mix-blend-mode: normal;`
`multiply`	Darkens by multiplying base color	`mix-blend-mode: multiply;`
`screen`	Brightens, used in duotones	`mix-blend-mode: screen;`
`overlay`	Combines multiply and screen (high contrast)	`mix-blend-mode: overlay;`
`darken`	Keeps the darker color	`mix-blend-mode: darken;`
`lighten`	Keeps the lighter color	`mix-blend-mode: lighten;`
`color-dodge`	Brightens the base color	`mix-blend-mode: color-dodge;`
`color-burn`	Darkens the base	`mix-blend-mode: color-burn;`
`difference`	Inverts colors based on difference	`mix-blend-mode: difference;`
`exclusion`	Similar to difference with lower contrast	`mix-blend-mode: exclusion;`
`hue`, `saturation`, `color`, `luminosity`	Use for artistic effects, image filtering	`mix-blend-mode: hue;`

Clip Path

It lets you define a visible area of an element using shapes or custom SVG paths. It hides everything outside the defined path.

Function / Shape	Use Case	Example Code
`circle()`	Profile pictures, buttons	`clip-path: circle(50%);`
`ellipse()`	Decorative image shapes	`clip-path: ellipse(50% 30%);`
`polygon()`	Diagonal sections, card angles	`clip-path: polygon(0 0, 100% 0, 100% 80%, 0 100%);`
`inset()`	Simple cropping	`clip-path: inset(10% 20%);`
`path('...')`	Complex SVG paths, blobs	`clip-path: path('M10,10 Q100,100 ... Z');`
`url(#clipPath)`	Advanced SVG clipping	`clip-path: url(#myCustomShape);`

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Html

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Crypto Wallet</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <div class="blob"></div>

  <div class="card-wrapper">
    <div class="card">
      <div class="glow-overlay"></div>
      <div class="content">
        <div class="title">NEON WALLET</div>
        <div class="balance">Ξ 3.96 ETH</div>
        <div class="address">0xEsh...1e13</div>
      </div>
    </div>
  </div>
</body>
</html>
```

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CSS

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}
body {
  font-family: 'Segoe UI', sans-serif;
  height: 100vh;
  background: linear-gradient(135deg, #090e1c, #1b1b3a, #2d1b4d);
  display: flex;
  align-items: center;
  justify-content: center;
  overflow: hidden;
}
.blob {
  position: absolute;
  width: 500px;
  height: 500px;
  background: radial-gradient(circle at 30% 30%, #ff00c8, #00fff0);
  filter: blur(130px);
  mix-blend-mode: screen;
  animation: moveBlob 18s ease-in-out infinite;
  z-index: 0;
}

@keyframes moveBlob {
  0%, 100% {
    transform: translate(-70px, -50px) scale(1);
  }
  50% {
    transform: translate(90px, 70px) scale(1.25);
  }
}

.card-wrapper {
  perspective: 1200px;
}
.card {
  position: relative;
  width: 350px;
  height: 200px;
  backdrop-filter: blur(18px);
  background: rgba(255, 255, 255, 0.05);
  border: 1px solid rgba(255, 255, 255, 0.15);
  border-radius: 20px;
  clip-path: polygon(0 0, 100% 0, 100% 80%, 90% 100%, 0 100%);
  box-shadow: 0 4px 30px rgba(0, 0, 0, 0.6);
  overflow: hidden;
  transition: transform 0.4s ease-in-out, box-shadow 0.3s ease;
  transform-style: preserve-3d;
  z-index: 1;
}

.card:hover {
  transform: rotateY(25deg) rotateX(5deg) scale(1.05);
  box-shadow: 0 60px 80px rgba(0, 0, 0, 0.7), 0 0 30px #00ffe0;
    -webkit-text-fill-color: rgb(23, 18, 58);
```

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CSS

```
.card:hover {
  transform: rotateY(25deg) rotateX(5deg) scale(1.05);
  box-shadow: 0 60px 80px rgba(0, 0, 0, 0.7), 0 0 30px #00ffe0;
    -webkit-text-fill-color: rgb(23, 18, 58);
}
.glow-overlay {
  position: absolute;
  top: 0;
  left: 0;
  width: 100%;
  height: 100%;
  background: linear-gradient(135deg, rgba(0,255,255,0.2), rgba(255,0,255,0.2));
  mix-blend-mode: overlay;
  pointer-events: none;
}

.content {
  position: relative;
  padding: 20px;
  color: #fff;
  z-index: 2;
}
.title {
  font-size: 1.3rem;
  font-weight: 600;
  background-clip: linear-gradient(90deg, #ff00ff, #00fff0);
  -webkit-background-clip: text;
  text-shadow: 0 0 8px rgba(255, 0, 255, 0.4);
  margin-bottom: 14px;
}

.balance {
  font-size: 2.4rem;
  font-weight: bold;
  color: #ffffff;
  text-shadow: 0 0 6px #0ff, 0 0 12px #00f0ff;
}

.address {
  font-size: 0.85rem;
  margin-top: 14px;
  color: #ccc;
  text-shadow: 0 0 2px rgba(255, 0, 255, 0.3);
}
```